

Manual

Statistical Process Analysis (MOC) PDV-SPA 8.1

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1 Statistische Prozessanalyse - Überblick

Summary

Possible fields of application

This function package includes the statistical reports/evaluations. In this context, samples are generated from the raw data on the basis of the sample parameters defined within the collection rules or their characteristics. To present the mean values of samples as graphic curve over time ideally complements the measured value curves of the graphic process analysis to represent the data set in a statistically smoothed way. Quality control charts enable statistical process monitoring and analysis on the basis of process values.

Integration

A machine interface for data collection in the HYDRA Process Communication Controller (PCC) as well as the licenses for process data collection and processing (PDV-PDM and PDV-VRP) are required for using this function package.

Functions

Possible analyses/evaluations of the statistical process analysis:

- Enhanced graphic process evaluation: Evaluation of samples in correlative process analysis. Mean values of samples and process limits are displayed for the samples in graphic form.
- Enhanced graphic process evaluation: Samples are evaluated on the basis of control charts using extensive filter criteria. Different control charts are possible (e.g. $\bar{x}$, s and R).

2 Graphic Process Analysis with Samples (based on machines)

Summary

Menu	Quality management → Process analysis → Graphic process analysis with samples (based on machines)
Transaction code	gpdasm
Function authorization	gpdasm

The graphic process analysis is used for the process analysis in quality management. The collection of samples enables the values to be compressed and allows for statistical preparation. The analysis function based on samples represents average values and their progression; it is an Xq chart.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. The selected process parameters can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily. In this evaluation, which refers to samples, only the averages from the samples are displayed. The collection of samples is defined in the collection request (inspection planning).

Integration

The function displays sample values from recorded measuring values of process characteristics. These values must have been previously collected in the system.

Requirement

The data compression (sampling) is active as a separate process.

Selection parameters

The following selection criteria are available in the application:

Workplace

The required workplace can be selected specifically in the "general" index tab using the pool of workplaces application.

Time domain from – until

Specifies a specific period of time for the evaluation.

Consider long-term data

Enabling this checkbox allows for long-term data to be considered.

Process parameters

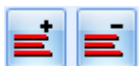
The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons



can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifying glass with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale”

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

3 Graphic Process Analysis with Samples (based on orders)

Summary

Menu	Quality Management → Process Analysis → Graphic Process Analysis with Samples (based on orders)
Transaction code	gpdaso
Function authorization	gpdaso

The graphic process analysis is used for the process analysis in quality management. The collection of samples enables the values to be compressed and allows for statistical preparation. The analysis function based on samples represents average values and their progression; it is an Xq chart.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. The selected process parameters can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily. In this evaluation, which refers to samples, only the averages from the samples are displayed. The collection of samples is defined in the collection request (inspection planning).

Integration

The function displays sample values from recorded measured values of process characteristics. These values must have been previously collected in the system.

Requirement

The data compression (sampling) is active as a separate process.

Selection parameters

The following selection criteria are available in the application:

MES order number

Selects a specific operation (MES order number)

Batch

Selects a batch number

Time domain from – until

Selects a specific period of time

Consider long-term data

Enabling this check box allows for long-term data to be considered.

Process parameters

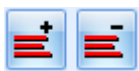
The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail applications

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The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

4 Graphic Process Analysis with Samples (based on batches)

Summary

Menu	Quality Management → Process analysis → Graphic process analysis with samples (based on batches)
Transaction code	gpdasc
Function authorization	gpdasc

The graphic process analysis is used for the process analysis in quality management. The collection of samples enables the values to be compressed and allows for statistical preparation. The analysis function based on samples represents average values and their progression; it is an Xq chart.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. The selected process parameters can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily. In this evaluation, which refers to samples, only the averages from the samples are displayed. The collection of samples is defined in the collection request (inspection planning).

Integration

The function displays sample values from recorded measured values of process characteristics. These values must have been previously collected in the system.

Requirement

The data compression (sampling) is active as a separate process.

Selection parameters

The following selection criteria are available in the application:

Batch number

Specifically selects a produced batch.

Alternative batch number 1

Selects the alternative batch number 1 at the produced batch

Time domain from – until

Selects a period of time

Consider long-term data

Enabling this check box allows long-term data to be considered.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons



can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

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